Neural Network Fitting Results

Best Stage-3 refinement, ablation comparison, and trajectory-level failure modes

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Evaluation Target

Source of this result set

The inference script loads the best refined student model and evaluates point-wise penetration mean and predictive standard deviation on the `series_wide_clean` trajectories.

Item	Setting
Best refinement run	`distill_cdf_onset_v2_loss_tuned_conservative_20260418_114747`
Teacher run	`stage2_engineered_nll_a_only_20260410_020607`
Evaluation split	`clean`, time window 0 ms to 5 ms
Raw/KD weights	reliable 0.95/0.06, uncertain 0.50/0.25, teacher-only 0.00/0.75
Shape timing	anchor/onset 0.025 ms; d2 gate start/transition 0.4 ms/0.2 ms

Raw result folder:
`rmse_eval_clean_20260418_115700_distill_cdf_onset_v2_loss_tuned_conservative_20260418_114747`

Data Scale and Best Overall Error

Metric	Value
CSV files	266
Trajectories	14,972
Prediction points	399,738
Truth range	0.00 mm to 95.88 mm
RMSE	6.60 mm
MAE	4.81 mm
Bias	−0.40 mm
Median absolute error	3.52 mm
P90 / P95 absolute error	10.92 mm / 13.91 mm
Range-normalized RMSE	6.88%

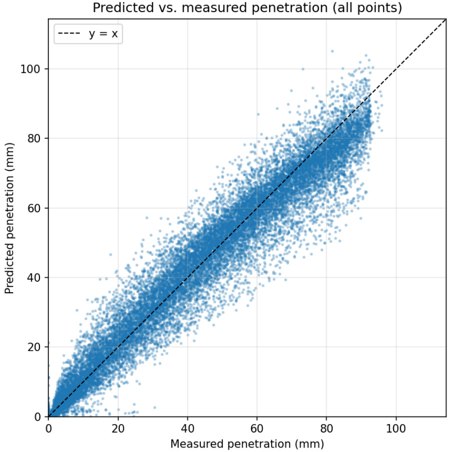
Change from the 4/16 baseline

- RMSE improves: 6.90 mm $\rightarrow$ 6.60 mm.
- MAE improves: 5.16 mm $\rightarrow$ 4.81 mm.
- Under-prediction is much smaller: bias -1.29 mm $\rightarrow$ -0.40 mm.
- Uncertainty coverage also improves: $1\sigma/2\sigma =$ 73.56%/96.93%.

Uncertainty

- Mean predicted σ: 6.40 mm
- 1σ coverage: 73.56%
- 2σ coverage: 96.93%

Predicted vs. Measured Penetration

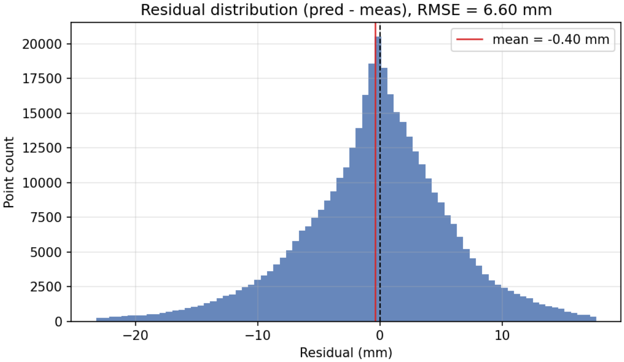


Main observations

- Most points still follow the $y = x$ line after loss tuning.
- The systematic low-bias problem is visibly reduced compared with the baseline run.
- Tail scatter remains in high-penetration and high-variance conditions, so trajectory-level review is still needed.

Each point is one time sample. The axes are measured and predicted penetration.

Residual Distribution and Calibration

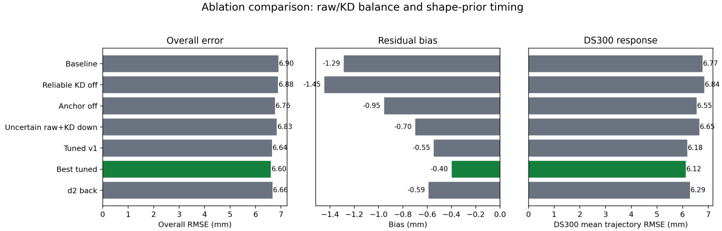


Residual definition

$$e = \hat{y}_{\mathrm{pen}} - y_{\mathrm{pen}}$$

- Bias $= -0.40\,\mathrm{mm}$; the global under-prediction is much weaker than before.
- 1σ coverage $= 73.56\%$, slightly conservative relative to the Gaussian reference.
- 2σ coverage $= 96.93\%$, also conservative but useful for screening high-risk trajectories.
- Calibration improved together with the mean-error metrics.

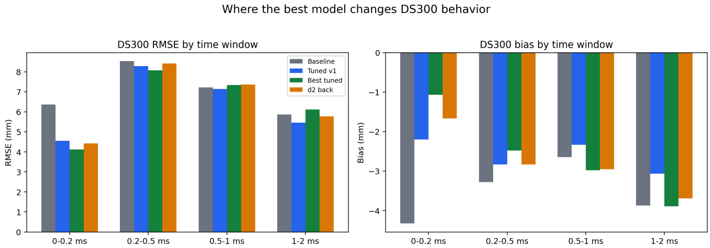
Ablation Comparison



Run	RMSE	Bias
Baseline	6.90	-1.29
Reliable KD off	6.88	-1.45
Anchor off	6.76	-0.95
Uncertain raw+KD down	6.83	-0.70
Tuned v1	6.64	-0.55
Best tuned	6.60	-0.40
d2 back	6.66	-0.59

- Lowering KD in reliable raw bins fixes the main bias direction.
- Combining raw/KD reweighting with shorter onset anchor gives the best overall fit.
- Moving the d2 gate back helps some HZ nozzles but hurts DS300 and overall RMSE.

Where the Best Run Changes DS300

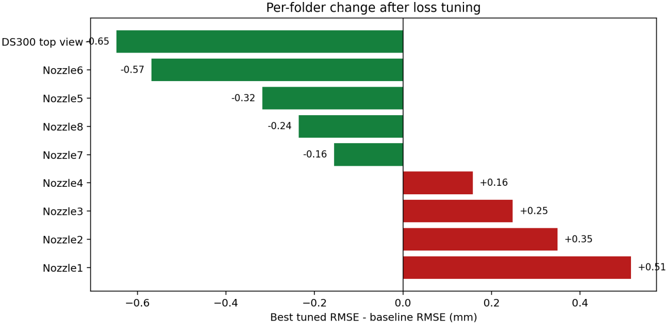


DS300 response

- DS300 mean RMSE: 6.77 mm $\rightarrow$ 6.12 mm.
- DS300 MAE: 5.99 mm $\rightarrow$ 5.25 mm.
- DS300 bias: -3.36 mm $\rightarrow$ -2.35 mm.

- The largest gain is at 0 ms to 0.2 ms, where the early-SOI under-prediction is reduced.
- The 0.5 ms to 2.0 ms region shows the remaining trade-off in curvature regularization.

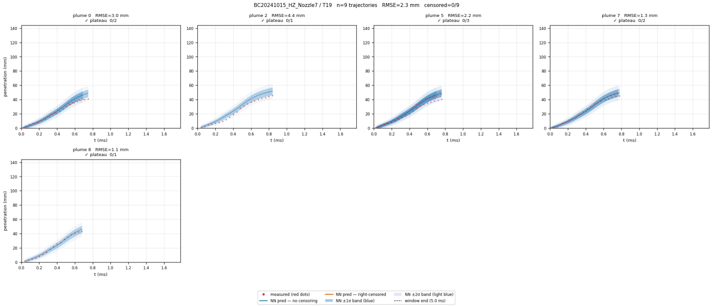
Per-Folder Trade-Off After Loss Tuning



Folder	Best RMSE
Nozzle7	3.52 mm
Nozzle5	3.63 mm
Nozzle6	4.65 mm
Nozzle8	4.81 mm
Nozzle4	5.40 mm
Nozzle1	5.73 mm
DS300 top view	6.12 mm
Nozzle2	6.75 mm
Nozzle3	9.21 mm

- The best run improves DS300, Nozzle5/6/7/8 most clearly.
- Nozzle1/2/3/4 reveal a cross-nozzle trade-off, likely tied to geometry-dependent early acceleration.

Best Case: Nozzle7 / T19

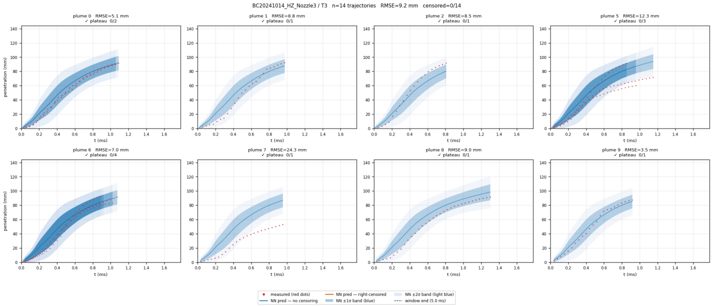


Representative characteristics

- Folder mean RMSE: 3.52 mm
- Best trajectory in this group: RMSE 0.47 mm
- 1σ / 2σ folder coverage: 85.63% / 97.93%

- The mean trend and plume-to-plume spread are captured well.
- The uncertainty band is slightly conservative but useful.
- This is the cleanest example of the tuned loss acting as a stable refinement rather than over-correction.

Worst Case: Nozzle3 / T3



Risk signals

- Folder mean RMSE: 9.21 mm
- Worst trajectory: RMSE 24.33 mm
- Worst-trajectory 1σ coverage: 8.3%

- The global loss tuning does not remove all nozzle-family-specific failure modes.
- Nozzle3 remains the hardest folder after tuning.
- This motivates geometry-aware priors or nozzle-conditioned loss scheduling.

Conclusions

1. The tuned Stage-3 run is the current best: RMSE is 6.60 mm, MAE is 4.81 mm, and NRMSE is 6.88%.
2. The main systematic issue is reduced: global bias improves from -1.29 mm to -0.40 mm; DS300 bias improves from -3.36 mm to -2.35 mm.
3. The ablation supports the KD-dominance hypothesis: strong KD in deterministic raw-supported bins pushes the student toward teacher bias; lower KD and restored raw supervision improve DS300.
4. The remaining errors are geometry-dependent: Nozzle3 and some HZ nozzles show a trade-off with DS300, suggesting early-SOI acceleration and curvature priors should be nozzle-aware.

Thesis-ready message

The loss tuning is not merely numerical hyperparameter search: it reveals a supervision conflict between raw deterministic observations and teacher distillation, and it exposes nozzle-family-dependent early spray acceleration behavior.

File Index

Content	File
Slides source	`Thesis/neural_network_fit_results_slides_en.tex`
Compiled PDF	`Thesis/neural_network_fit_results_slides_en.pdf`
Copied/generated figures	`Thesis/images/neural_network_fit_results/`
Best result folder	`rmse_eval_clean_20260418_115700_distill_cdf_onset_v2_loss_tuned_c`
Ablation CSV	`ablation_summary_for_slides.csv`
DS300 time-bin CSV	`ds300_time_bins_for_slides.csv`